

EX1 $V_0 = 10.5V$
 $R_{EZ} = ?$
 $P = ?$

$$R_{AB} = R_{B2} // R_{A1}$$

$$= 68.71 \text{ K}\Omega$$

$$\frac{V_o}{R_L} = I_{Q2} = 1.875 \text{ mA}$$

$$V_{a1} - R_{a2} \frac{I_{a1}}{\beta F} - V_{a2} - R_{a1} \frac{I_{a1}}{\beta F} \left(1 + \frac{1}{\beta F}\right) - R_{a1} \frac{I_{a2}}{\beta F a} = 0$$

$$I_a = V_B - V_{BE1} - R_{E1} \frac{I_{C2}}{\beta_{F2}} = 3.246 \text{ mA}$$

$$\frac{P_{BB}}{BF_1} + P_{B1}\left(\frac{1}{BF_1}\right)$$

$$(I_{L1}, V_{L1}) = (3.246 \text{ mA}, 11.23 \text{ V})$$

$$(I_{I_2}, V_{CE_2}) = (1.875 \text{ mA}, -2.27 \text{ V})$$

$$R_{ED} = 5.95 \text{ K}\Omega$$

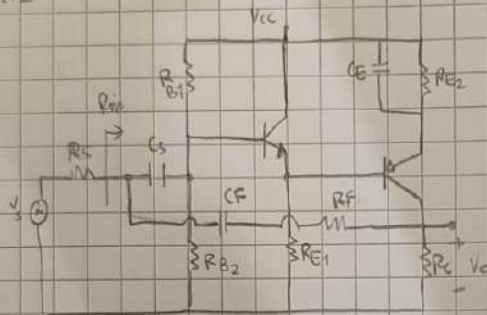
$$V_{KE2} = V_0 = \left[R_{E1} \left(I_{Q1} \left(1 + \frac{1}{\beta_{F1}} \right) + I_{Q2} \right) + V_{BE2} \right]$$

$$= -2.27 \text{ V}$$

$$V_{CE1} = V_{CC} - R_{E1} \left[I_{C1} \left(1 + \frac{1}{\beta_{F1}} \right) + I_{B2} \right]$$

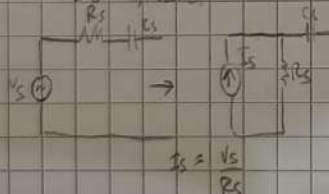
$$= 19.23 \text{ V}$$

$$R_{F2} = V_{G2} - \left(V_0 - V_{(F2)} \right) = 5.95 \text{ k}\Omega$$



Feedback topology: current mixing - voltage sampling
(shunt) (shunt)

Not an equivalent



RF $\rightarrow$ 100V $\rightarrow$ 3A

$$\begin{aligned} G_F &= 2F \\ G_L &= RF \end{aligned}$$

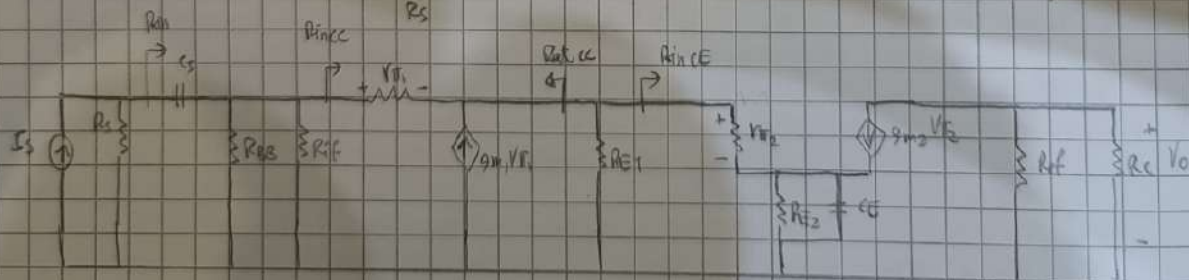
$$\beta = \frac{dI_1}{dV_1} \bigg|_{V_2=0} = -\frac{1}{R_F}$$

$$q_{m_2} = \frac{I_a}{V_T} = 0.125 \text{ A}^{-1}$$

$$P_{A2} = \frac{I_{A2}}{V_T} = 0.072 \text{ W}$$

$$r_{\pi_1} = \beta_{01} = 1.6 \text{ k}\Omega$$

$$R_{S2} = \frac{\beta_{02}}{\alpha_2} = 2.08 \text{ k}\Omega$$



$$\dim C^E = r_{H_2}$$

$$P_{\text{in CE}} = P_{\text{in}} + (1 + \beta_{\text{DC}})(R_{E3} \parallel R_{\text{in IE}}) = 276.26 \text{ mW}$$

$$A^2 = \frac{N_0}{15} = \alpha; \alpha_{CC} \cdot \alpha_{CE}$$

$$\alpha_{CE} = -g_{m2}(R_{of} \parallel R_C) = -258.5 \text{ [V/V]}$$

$$d_{cc} = \frac{(1 + \beta_{01}) R_E \parallel R_{inCE}}{R_{inCE}} = 0.994 \text{ V/V}$$

$$d_i = R_5 \parallel R_{B3} \parallel R_{iF} \parallel R_{inCE} = 852 \Omega$$

$$A_{a2} = -229,2 \text{ kN}$$

$$A_F = \frac{A_d}{1 + \beta A_d} = -9.582 \text{ k}\Omega$$

$$A_v = \frac{A_F}{R_s} = -9.582$$

$$R_{in} = R_3 \parallel R_{BB} \parallel R_{if} \parallel R_{inC} = 892 \Omega$$

$$R_{in}^F = \frac{R_{in}^A}{1 + \beta A_{vcl}} = 37.3 \Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}'} - \frac{1}{R_s}} = 38.75 \Omega$$

Considering only CE as effective C_1, C_2 started

$$R_{in} = \frac{1}{\frac{1}{R_{in}'} + \frac{1}{R_s}}$$

$$R_{in}^{SC} = R_{in} \parallel Z_x$$

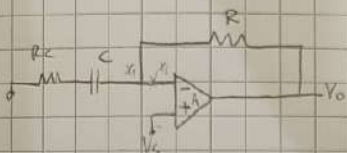
$$R_{out} = \frac{V_{in} + R_s \parallel R_{in} / R_{in}}{1 + \beta_{02}} = 12.6 \Omega$$

$$Z_x = 13.8 \Omega$$

$$R_{in}^{SC} = 13.76 \Omega$$

$$f_L = 11.56 \text{ kHz} \rightarrow f_H = \frac{f_L}{1 + \beta_{02}} = 483 \text{ Hz}$$

Ex 2



$$A(s) = \frac{A_{dc}}{1 + \frac{s}{\omega_0}}$$

$$A_{dc} = 10^5$$

$$\omega_0 = 500 \text{ rad/s}$$

$$R = 10 \text{ k}\Omega$$

$$C = 10 \text{ nF}$$

considering $R_c = 0 \Omega$

1) non inverting configuration $\frac{V_o}{V_s} = A_v(s) = 1 + \frac{R}{R_{in}^{SC}} = 1 + \frac{sRC}{1 + \beta_{02}}$

$$2) T = \frac{A_{dc}}{1 + \frac{s}{\omega_0}} = \frac{A_{dc}}{1 + \frac{sRC}{1 + \beta_{02}}}$$

$$\beta = \frac{1}{1 + \beta_{02}}, \quad \omega_{RC} = \frac{1}{RC} = 10 \text{ krad/s}$$

$$\frac{1}{\beta} = 1 + \beta_{02} = 1 + \frac{s}{\omega_0}$$

when $\omega \ll \omega_{RC} \rightarrow \frac{1}{\beta} = 0 \text{ dB}$ (voltage divider, C opened)

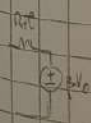
$$A_{dc} \omega_0 = A_{dc} \omega_{RC} \beta$$

$$\frac{1}{\beta} = \frac{A_{dc}}{\omega_{RC}} \Rightarrow A_{dc} = \frac{\omega_{RC}}{\beta}$$

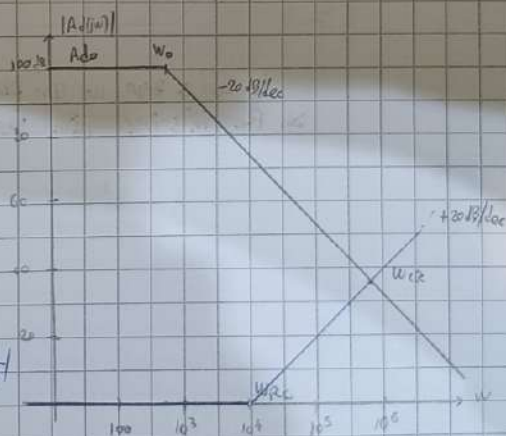
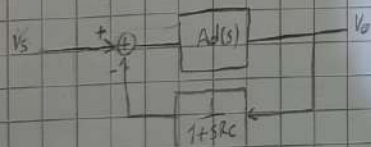
$$\omega_{RC} = \sqrt{A_{dc} \omega_0 \beta} = 70.7 \text{ krad/s}$$

$$PM = 180 - \angle \left(\frac{\omega_{RC}}{\omega_0} \right) - \angle \left(\frac{\omega_{RC}}{\omega_{RC}} \right) = 0.85^\circ \leftarrow \text{low phase margin}$$

3) voltage divider, voltage sampling $\rightarrow$ voltage amplifier



For non inverting configuration block diagram is the same using direct loop or feedback theory



4) maximum value of R_c same frequency response up to ω_{CR}

$$\text{considering } R_c \rightarrow \beta = \frac{1 + R_c}{sC} = \frac{1 + sRC}{(R + R_c)sC + 1} \rightarrow \omega_z = \frac{1}{RC}$$

$$\omega_p = \frac{1}{(R + R_c)C} \quad \text{to maintain the same behavior up to the } \omega_{CR} \quad R_c \ll R \quad \text{so that } \omega_p \approx \omega_{CR} \quad \text{and } \omega_z \approx \omega_{CR}$$

$$\omega_z = \omega_{CR} \rightarrow R_c = \frac{1}{\omega_{CR}C} = 141 \Omega$$

keeping $R_c < 1k\Omega$ allows to keep the ω_{CR} at the right, or at least exactly over ω_{CR} . Here over $R_c \ll R$ is verified and ω_{CR} is not moved too much so everything is unchanged with ω_{CR}

$$R_c = 141 \Omega$$

$$\omega_z = \omega_{CR} \rightarrow \text{AP} = +45^\circ$$

$$\omega_p = \frac{1}{R + R_c} = 9.86 \text{ Krad/s}$$

$$\rightarrow \text{PT} = 180^\circ - \arg\left(\frac{\omega_{CR}}{\omega_z}\right) - \arg\left(\frac{\omega_{CR}}{\omega_p}\right) + \arg\left(\frac{\omega_{CR}}{\omega_z}\right) = 15.94^\circ$$

or equivalently

$$\text{we have to add a zero in } \beta = \frac{1}{sC} = \frac{1}{\frac{1}{sC} + R} = \frac{1}{1 + sRC}$$

in some other ways

$$\text{For example placing a cap } \parallel \text{ to } R \rightarrow \beta = \frac{1}{sC} = \frac{1}{\frac{1}{sC} + (R \parallel \frac{1}{sC})}$$

10 e 30 min

$$\beta = \frac{1}{sC} = \frac{1 + sCR}{1 + sR(C + C)} = \frac{1 + sCR}{1 + sR(C + C)}$$

equivalent approach

to add a zero we can place
a R_c in series with the cap C or a C_c cap in parallel with the R resistor.

$$\omega_{CR} = 1 \rightarrow C_c = \frac{1}{R\omega_{CR}} = 14.14 \text{ nF}$$

